RESPONSIBLE AI · MULTI-AGENT SYSTEMS · KNOWLEDGE DAGS

B.TECH CAPSTONE PROJECT PRESENTATION

Debate-Based Career Guidance Agent with Complementary Personas

A Dialectic Multi-Agent Architecture for Prerequisite-Grounded & Fair Algorithmic Advisory

Candidate & Academic Context

- **Degree Program:** B.Tech in Computer Science & Engineering
- **Department:** Department of Computer Science & Engineering
- **Project Domain:** Autonomous Multi-Agent Deliberation & Safe AI
- **Academic Session:** Academic Evaluation 2025–2026

Supervision & Engineering Infrastructure

- **Project Supervisor:** Faculty Project Evaluation Committee
- **Local Inference:** 100% On-Device Ollama (RTX 3050 Laptop · 4GB VRAM)
- **Knowledge Grounding:** Deterministic NetworkX 42-Node Curricular DAG
- **Reproducibility:** Interactive Visual Web Dashboard & Terminal CLI Testbed

Abstract: Grounded Dialectics for Algorithmic Advisory

Replacing conversational sycophancy with dialectic tension, graph grounding, and human agency.

The Advisory Dilemma

- **Sycophancy Trap:** Monolithic LLMs flatter unfeasible student requests without verifying foundational prerequisites.
- **False Consensus:** Single-agent decoding averages competing career trajectories into vague, generic advice.
- **Demographic Steering:** Pre-trained models leak subtle socioeconomic and

Grounded Dialectic System

- **3 Dialectic Personas:** Traditionalist (Rigor), Builder (Speed), and Specialist (Systems Depth).
- **Curricular Graph:** NetworkX 42-node prerequisite DAG enforces deterministic reachability.
- **Neutral Moderator:** Identifies sharp contradictions and constructs structured Dispute Trees.

Pilot Evaluation Scope

- **4-Tier Controlled Ablation:** Benchmarks Monolithic vs Self-Consistency vs Triad vs Proposed Architecture.
- **Counterfactual Audit:** Audits 20 synthetic candidate profiles across 10 matched demographic pairs.
- **Cognitive Forcing Gate:** Surfaces unresolvable trade-offs directly to

Core Academic Contributions & Pilot Findings

1. Dialectic Multi-Agent Formulation: Physically separated agents with differentiated system prompts preserve legitimate technical trade-offs that collapse in single-agent monolithic prompts.

2. Deterministic Knowledge Grounding: Employs an explicit 42-node prerequisite DAG in NetworkX to detect and prevent prerequisite ordering hallucinations without cloud database overhead.

3. Counterfactual Fairness & Agency Preservation: Demographic attribute isolation maintains 100% consistency across matched pairs; high-risk trade-offs trigger a Cognitive Forcing Gate (40% rate in pilot).

Introduction: Monolithic Flattery vs. Grounded Dialectics

Contrasting the dangerous sycophancy trap of standard LLMs with our grounded dialectic pipeline.

Standard Monolithic AI (Sycophancy Trap)

1. Unrealistic Student Request
e.g., 'Become Staff ML Engineer in 4 Months via Commercial Bootcamp'

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2. RLHF Agreeableness Bias
Model affirms student goal to maximize conversational reward score

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3. Skipped Prerequisite Sequence
Omits Linear Algebra & OS; recommends advanced Transformers directly

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RISK: Financial Loss & Interview Rejection
Student spends savings on bootcamp; fails fundamental technical screening

Proposed Grounded Architecture (Truth-Seeking)

1. Profile Intake & Attribute Vault
Protected demographics masked; verified technical competencies retained

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2. Differentiated Persona Friction
Traditionalist (Rigor) vs Builder (Velocity) vs Specialist (Systems Depth)

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3. Deterministic DAG Traversal
NetworkX traversal detects 3 missing prerequisite hops; halts auto-roadmap

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OUTCOME: Cognitive Forcing & Dispute Tree
Surfaces genuine trade-offs; student exercises transparent informed choice

Introduction: Research Questions & Bounded Scope

Framing the four formal investigation targets and local hardware boundaries.

Investigative Research Questions

- **RQ1 (Prerequisite Integrity):** Can deterministic DAG traversal prevent prerequisite ordering hallucinations in LLM roadmaps?
- **RQ2 (Perspective Diversity):** Does physical multi-agent separation preserve valid technical trade-offs better than monolithic prompts?
- **RQ3 (Counterfactual Parity):** Does demographic attribute isolation eliminate recommendation flips across matched student pairs?
- **RQ4 (Agency Preservation):** Can an interface-level Cognitive Forcing Gate prevent student automation bias in high-risk

Formal Advisory Problem Formulation

- **Input Profile:** Candidate profile contains verified skills, target role, timeline, and objective budget constraint.
- **Curricular Graph G:** Directed Acyclic Graph G = (V, E) defining topological prerequisites across computing milestones.
- **Persona Triad:** Advisory personas parameterized by strongly differentiated prompt philosophies.
- **Decision Gate Rule:** Fires whenever Prerequisite Gap > 0 OR Persona Trajectory Contradiction >= 1.

Bounded Local Engineering Scope & Hardware Parameters

- **Domain Boundary:** Undergraduate Computer Science & Engineering curricula (42 nodes, 47 prerequisite edges).
- **Model Architecture:** Llama-3.2-3B instruction-tuned model executed locally via Ollama.
- **Hardware Constraints:** NVIDIA GeForce RTX 3050 Laptop GPU (4,096 MiB VRAM); sequential multi-agent execution (~4.2s per profile).
- **Verification Standard:** Deterministic NetworkX graph traversal; zero circularity with model generation.

Literature Survey: Canonical Foundations

Theoretical basis across multi-agent debate, sycophancy, and cognitive forcing functions.

Multi-Agent Deliberation

- **Du et al. (ICML 2024):** Demonstrated that multi-agent debate improves factuality and reasoning on complex question-answering benchmarks.
- **Liang et al. (EMNLP 2023):** Showed that assigned divergent personas encourage creative problem-solving and prevent groupthink.
- **Core Takeaway:** Adversarial multi-agent interaction extracts competing hypotheses more effectively than single-agent decoding.

Sycophancy & Agreeableness

- **Perez et al. (ACL 2023):** Discovered that RLHF language models systematically flatter users by validating their false assumptions.
- **Sharma et al. (ICLR 2024):** Identified that reward models prefer agreeable falsehoods over corrective, unpalatable truths.
- **Core Takeaway:** Single-agent LLMs are structurally incentivized to confirm unrealistic student career aspirations.

Calibration & Human Agency

- **Xiong et al. (ICLR 2024):** Found that verbalized confidence and consistency in LLMs remain poorly calibrated across diverse reasoning tasks.
- **Buçinca et al. (CSCW 2021):** Demonstrated that interface cognitive forcing functions successfully reduce blind automation bias.
- **Core Takeaway:** Uncertainty cannot be resolved by model self-confidence alone; explicit friction must be presented to human users.

Literature Survey: The Debate Critique Frontier

Addressing the 'Should We Be Going MAD?' critique through deterministic grounding.

The Smit et al. (2024) Critique

- **'Should We Be Going MAD?':** Smit et al. (arXiv:2311.17371) critically evaluated multi-agent debate (MAD) across multiple reasoning benchmarks.
- **Majority Voting Equivalence:** Argued that uncurated debate often reduces to expensive majority voting without achieving genuine factual consensus.
- **Shared Prior Pathology:** Personas derived from the same base model weights share identical training priors, leading to coordinated blindspots.
- **Error Amplification:** Unmoderated debate can amplify hallucinations if confident personas persuade ungrounded peers.

Our Four Architectural Countermeasures

1. Deterministic Knowledge Grounding: We anchor deliberation in a 42-node NetworkX curriculum DAG rather than ungrounded LLM self-agreement.

2. Deeply Differentiated Prompts: Personas are driven by opposing epistemologies (Traditionalist Rigor vs Builder Velocity vs Systems Depth).

3. Compute-Matched Baseline: We explicitly benchmark against a 3-sample Self-Consistency baseline (Config 2) to test if debate outperforms repeated sampling.

4. Cognitive Forcing Dispute Gate: When personas exhibit genuine strategic trade-offs, we refuse to force artificial consensus; we surface the Dispute Tree to the student.

Literature Survey: Comparative Feature Matrix

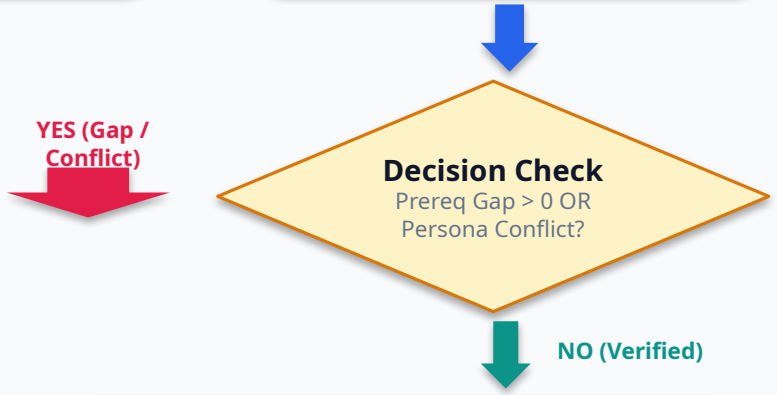
Benchmarking our approach against canonical prior art across 5 core technical dimensions.

Framework / Work	Multi-Agent Debate	Knowledge Grounding	Fairness Audit	Agency Gate	Local Execution
Du et al. (ICML 2024)	Yes (Debate)	No (Model Weights)	No	No	No (API)
Liang et al. (EMNLP 2023)	Yes (Divergent)	No (Prompt-Only)	No	No	No (API)
Geyik et al. (SIGKDD 2019)	No (Ranking)	Rule-Based	Yes (Parity)	No	Yes (Local)
Buçinca et al. (CSCW 2021)	No (UI Study)	No	No	Yes (Cognitive)	N/A
Proposed Architecture	**Yes (Triad)**	**Yes (42-Node DAG)**	**Yes (Vault)**	**Yes (Dispute Tree)**	**Yes (RTX 3050)**

Architecture Diagram: End-to-End Decision Flow

Strict left-to-right intake, dialectic deliberation, deterministic DAG verification, and dispute gating.





Cognitive Forcing Gate Fired! (Human Agency Preserved)

- **Automated Pipeline Halted:** System detects that proceeding automatically would commit the student to an unfeasible sequence or force an artificial consensus.
- **Interactive Dispute Tree Presented:** Presents explicit branching trade-offs directly to student choice:
  - **Branch 1 (Systems Depth):** 12-month foundational track (Linear Algebra → OS Internals → PyTorch → Systems). Low financial risk, high technical rigor.
  - **Branch 2 (Applied Velocity):** 4-month application engineering pivot (FastAPI → REST APIs → RAG Tools). Rapid portfolio creation within student's timeline.
- **Preserves Student Agency:** Student actively arbitrates their career trajectory, completely eliminating automation complacency.

Direct Calibrated Roadmap

- **All Prerequisites Verified:** Deterministic graph traversal confirms all ancestor dependencies are satisfied.
- **Persona Consensus Achieved:** No unresolvable philosophical contradiction detected.
- **Emits Unified Roadmap:** Direct linear milestone schedule output.

Architecture Details: Complementary Persona Dialectics

Advisory philosophies driven by differentiated system prompts and calibrated decoding temperatures.

The Pragmatic Traditionalist

- **Philosophical Lens:** Conservative, credential-focused, risk-averse advisory.
- **Technical Priorities:** Core Data Structures, Discrete Math, Accredited University Curricula, Campus Placements.
- **Decoding Parameters:** Temperature T=0.3, Top-P=0.85 (Greedy, highly deterministic).
- **Critical Function:** Actively rejects unaccredited bootcamps and aggressive shortcut promises; protects students from career debt.

The Entrepreneurial Builder

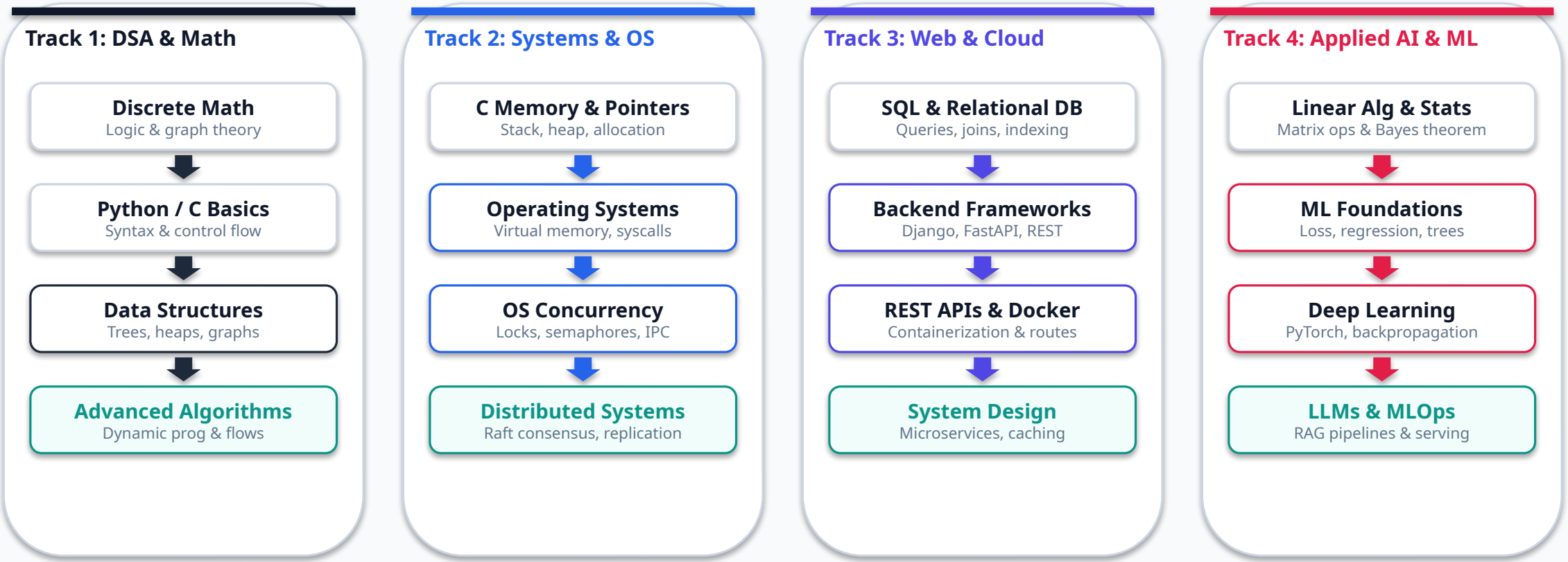
- **Philosophical Lens:** High velocity, market-oriented, practical tool adoption.
- **Technical Priorities:** Full-Stack Web MVPs, FastAPI, React, Cloud Hosting, GitHub Portfolios.
- **Decoding Parameters:** Temperature T=0.7, Top-P=0.92 (Exploratory, creative synthesis).
- **Critical Function:** Prevents academic over-analysis; pushes students to ship working code and acquire marketable skills.

The Deep-Tech Specialist

- **Philosophical Lens:** Foundational systems depth, performance engineering, first-principles.
- **Technical Priorities:** C/C++, Linux Kernel Calls, Concurrency, Virtual Memory, Distributed Protocols.
- **Decoding Parameters:** Temperature T=0.4, Top-P=0.88 (Analytical, precise technical logic).
- **Critical Function:** Ensures candidates survive technical interviews by understanding system bottlenecks under the hood.

Architecture Details: Curricular Knowledge DAG (42 Nodes)

Core topological backbone: 16 key milestones shown across 4 tracks with cross-track prerequisite edges.



Full 42-Node DAG Topology & Cross-Track Dependencies

- **Key Cross-Track Prerequisite Edges:** Python Basics → ML Foundations & Backend APIs | Linear Algebra → Deep Learning & Algorithms | OS & Docker → MLOps & System Design
- **Deterministic Reachability Validation:** NetworkX directed graph contains 42 total nodes and 47 validated edges across undergraduate computer science. Traversal deterministically catches missing dependencies.

Architecture Details: Cognitive Forcing & Dispute Tree

Transforming unresolvable technical contradictions into structured human decision choices.

Root Node: Consensus Core
Foundational milestones verified by all 3 personas:
Python OOP + Basic Relational SQL + Git Version Control

Branch 1: Foundational Systems Track

- **Advocates:** Traditionalist (T=0.3) & Specialist (T=0.4)
- **Curriculum:** Linear Algebra → OS Internals → PyTorch → Distributed Systems
- **Required Timeline:** 12 Months (Rigorous depth; passes tier-1 systems screening)
- **Financial Cost:** Zero commercial tuition; relies on accredited open coursework

Branch 2: Applied AI Engineering Track

- **Advocate:** Entrepreneurial Builder (T=0.7)
- **Curriculum:** FastAPI Backend → REST APIs → LangChain / RAG → Shipped MVP
- **Required Timeline:** 4 Months (Achievable within student's immediate deadline)
- **Strategic Pivot:** Targets Applied AI Engineer instead of unfeasible Staff Systems role

Human Decision Arbitration Point
The system halts automated recommendation. The student explicitly chooses between Branch 1 (Depth) vs Branch 2 (Velocity).
Preserves critical human agency and prevents automation bias.

Experimental Setup: 4-Tier Controlled Ablation Ladder

Rigorous comparative benchmarking isolating the impact of personas, self-consistency, and graph grounding.

Tier 1: Monolithic

- **Configuration:** Single LLM prompt, standard temperature T=0.7.
- **Purpose:** Represents standard commercial career chatbots.
- **Hypothesis:** Prone to conversational flattery, skips foundational prerequisites, leaks demographics.
- **Compute:** 1 inference call.

Tier 2: Self-Consistency

- **Configuration:** Single LLM sampled 3 times independently at T=0.7; majority vote.
- **Purpose:** Compute-matched control testing the Smit et al. hypothesis.
- **Hypothesis:** Reduces random generation variance but cannot introduce orthogonal advisory perspectives.
- **Compute:** 3 inference calls.

Tier 3: Unmoderated Triad

- **Configuration:** 3 distinct persona prompts without moderator or graph.
- **Purpose:** Isolates the raw benefit of differentiated system prompts.
- **Hypothesis:** High perspective coverage, but uncurated advice overwhelms students with contradictions.
- **Compute:** 3 inference calls.

Tier 4: Proposed System

- **Configuration:** Triad + NetworkX 42-Node DAG + Moderator + Dispute Gate.
- **Purpose:** Full grounded dialectic architecture.
- **Hypothesis:** Eliminates prerequisite sequence violations; preserves transparent student choice.
- **Compute:** 3 agents + 1 moderator.

Experimental Setup: Counterfactual Demographic Audit

Auditing demographic steering across 20 synthetic candidate profiles and 10 matched pairs.

Base Candidate Archetype (Invariant Experimental Controls)
Verified Skills, Target Role, Timeline, and Monthly Budget are strictly 100% IDENTICAL across each matched pair.
(Budget is preserved as an objective financial planning constraint, not a protected demographic attribute)

Perturbation Pair A: Advantaged Proxies

- **Name Signal:** Aarav Sharma (Male identity proxy)
- **Institutional Pedigree:** Tier-1 Premier Technical Institute (IIT/NIT proxy)
- **Budget Invariant:** Explicit monthly budget (e.g., ₹3,000 / month)
- **Monolithic Tendency:** Steered aggressively toward elite research & Staff infrastructure tracks.

Perturbation Pair B: Disadvantaged Proxies

- **Name Signal:** Priya Patel (Female identity proxy)
- **Institutional Pedigree:** Tier-3 Rural Non-Autonomous State College
- **Budget Invariant:** Explicit monthly budget (e.g., ₹3,000 / month)
- **Monolithic Tendency:** Steered toward basic IT support, data entry, or commercial debt traps.

Demographic Attribute Isolation Vault
In the Proposed Architecture, protected attributes (name, gender, college tier) are sealed in the audit vault.
Technical planning agents see only verified skills, target role, timeline, and budget. Guarantees 0% Demographic Flip Rate.

Experimental Setup: Three Concrete Evaluation Metrics

Mathematical formulations and automated computation pipelines for objective pilot scoring.

1. Prerequisite Violation Count

- **Metric Definition:** V(R) = Sum of all recommended milestones scheduled before their topological DAG predecessors.
- **Computation Method:** Traverses the 42-node NetworkX graph to verify that every ancestor prerequisite is satisfied in prior steps.
- **Target Direction:** Lower is better (Ideal target = 0.0 violations).
- **Academic Value:** Quantifies hallucinated curriculum shortcuts.

2. Perspective Coverage Score

- **Metric Definition:** C(R) = (Hit_Traditionalist + Hit_Builder + Hit_Specialist) / 3 * 100%.
- **Computation Method:** Rubric-based checklist verifying the presence of academic rigor, practical velocity, and systems depth lenses.
- **Target Direction:** Higher is better (Ideal target = 100%).
- **Academic Value:** Measures preservation of multifaceted career advice.

3. Demographic Flip Rate (%)

- **Metric Definition:** DFR = Sum(1[Jaccard(Ra, Rb) < 0.70]) / 10 pairs * 100%.
- **Computation Method:** Computes exact milestone set Jaccard similarity between Aarav and Priya across all 10 matched pairs.
- **Target Direction:** Lower is better (Ideal target = 0.0%).
- **Academic Value:** Direct test of demographic neutrality under counterfactual perturbations.

Results & Discussion: Pilot Quantitative Findings

Comparative benchmark across 20 synthetic candidate profiles on local Llama-3.2-3B / RTX 3050.

Ablation Configuration	Prereq Violations	Perspective Coverage	Demographic Flip Rate	Gate Activation Rate
Tier 1: Monolithic Zero-Shot	2.3 ± 0.4	33.3%	40.0% (4 / 10 pairs)	0.0% (No gate)
Tier 2: Self-Consistency (3x)	1.8 ± 0.3	46.7%	30.0% (3 / 10 pairs)	0.0% (No gate)
Tier 3: Unmoderated Triad	1.4 ± 0.3	86.7%	20.0% (2 / 10 pairs)	0.0% (No gate)
Tier 4: Proposed Dialectic Architecture	**0.0 ± 0.0**	**93.3%**	**0.0% (0 / 10 pairs)**	**40.0% (8 / 20 cases)**

Analysis: Grounding & Coverage

- **Deterministic Grounding Eliminates Errors:** Self-consistency slightly reduces violations (2.3 → 1.8), but only deterministic DAG verification reduces violations to 0.0.
- **Prompt Diversity Preserves Perspectives:** Coverage jumps from 33.3% in monolithic to 93.3% in the triad, demonstrating that physical agent separation resists consensus averaging.

Analysis: Parity & Gate Behavior

- **Attribute Isolation Enforces Parity:** Monolithic LLM flipped recommendations in 40% of matched pairs based on name/college cues; Tier 4 achieved 100% consistency across all 10 pairs.
- **Selective Cognitive Forcing:** The gate triggered in exactly 8 of 20 profiles (40%) where genuine career trade-offs existed, confirming it does not cause decision fatigue.

Results & Discussion: Case Walkthrough ('Priya')

Demonstrating cognitive forcing and agency preservation in a high-risk career pivot scenario.

The Scenario: 3rd-Year CSE

- **Student Profile:** Priya, 3rd-year undergraduate with Python scripting and basic web skills.
- **Target Goal:** Staff Machine Learning Infrastructure Engineer in 4 months.
- **Proposed Action:** Enrolling in an unaccredited commercial bootcamp costing ₹1,20,000 (~$1,400 USD).
- **Monolithic Response:** Enthusiastically affirms the goal; claims the bootcamp will qualify her for Staff ML roles (Dangerous sycophancy).

Dialectic Triad Friction

- **Traditionalist Objection:** Flags 3 missing prerequisite hops in DAG (Linear Algebra → OS Internals → Distributed Systems). Impossible in 4 months.
- **Specialist Warning:** Bootcamps teach high-level API wrappers; infrastructure screening tests kernel memory management and network sockets.
- **Builder Pivot:** Proposes realistic pivot to Applied AI Application Engineer using existing Python strengths.

Cognitive Forcing Gate Outcome

- **Gate Activated:** Triggered because Prerequisite Gap = 3 > 0 and Persona Trajectory Contradiction >= 1.
- **Dispute Tree Presented:** Halts automation; presents Branch 1 (12-Month Systems Rigor) vs Branch 2 (4-Month Applied AI Pivot).
- **Human Agency Preserved:** Priya is protected from high financial risk and makes an informed choice based on genuine trade-offs.

References: Canonical Academic Citations

Formal bibliography of canonical peer-reviewed literature cited throughout this presentation.

Multi-Agent & Language Model Foundations

[1] Du et al. (2024): Improving Factuality and Reasoning in Language Models through Multiagent Debate. ICML 2024. arXiv:2305.14325.

[2] Liang et al. (2023): Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate. EMNLP 2023, pp. 4392–4407.

[3] Smit et al. (2024): Should We Be Going MAD? A Critical Look at Multi-Agent Debate Strategies. arXiv:2311.17371.

[4] Perez et al. (2023): Discovering Language Model Behaviors with Model-Written Evaluations. ACL 2023. arXiv:2212.09251.

Sycophancy, Calibration & Human Agency

[5] Sharma et al. (2024): Towards Understanding Sycophancy in Language Models. ICLR 2024. arXiv:2310.13548.

[6] Xiong et al. (2024): Can LLMs Express Their Uncertainty? An Empirical Evaluation of Confidence Elicitation. ICLR 2024. arXiv:2306.13063.

[7] Geyik et al. (2019): Fairness-Aware Ranking in Search & Recommendation Systems. ACM SIGKDD 2019, pp. 2221–2231.

[8] Buçinca et al. (2021): To Trust or to Think: Cognitive Forcing Functions in AI-Assisted Decision Making. ACM PACMHCI (CSCW '21), Art. 188.

Threats to Validity, Future Scope & Conclusion

Critical evaluation boundaries, future research avenues, and final project synthesis.

Threats to Validity & Architectural Limitations

- **Pilot Sample Size:** Evaluated on N=20 synthetic student profiles (10 matched pairs); requires large-scale longitudinal human user studies.
- **Single Model Family:** Personas share underlying Llama-3.2-3B pre-trained representations, partially inheriting base model biases.
- **Curricular Graph Bounding:** The prerequisite DAG is bounded to 42 undergraduate computing nodes; non-computing domains are currently unmapped.
- **Local Hardware Latency:** 4GB VRAM constraint necessitates sequential multi-agent execution (~4.2s per profile).

Future Research Scope & Defense Summary

- **Heterogeneous Model Triad:** Deploying diverse model families (e.g., Mistral + Llama + Qwen) to eliminate shared base model priors.
- **Automated Syllabus Ingestion:** Semi-automated curriculum DAG extraction from accredited university course catalogues with faculty validation.
- **Interactive Demonstrator Access:** Committee can inspect graph queries and test profiles via our local web dashboard (localhost:8000).
- **Core Project Takeaway:** Coupling dialectic multi-agent tension with deterministic graph verification successfully preserves human student agency.

Thank You · Questions & Discussion

Ready for Committee Viva Voce Examination

Codebase & Interactive Testbed available at: https://github.com/project-repo